DATASET DETAILS

Online Retail II — Retail Transaction Analytics Platform

Dataset inventory, schema, preprocessing, quality findings, and known analytical considerations.

Project input file
`data/raw/Online_Retail_II.xlsx`

Processed output
`data/processed/cleaned_retail.parquet`

1. Dataset Identification

Item	Details
Dataset name	Online Retail II
Local source file	data/raw/Online_Retail_II.xlsx
Workbook sheets	Year 2009-2010; Year 2010-2011
Format	Excel workbook (.xlsx)
Project usage	Historical retail transaction analysis, sales summaries, product and customer analytics.
Public source	UCI Machine Learning Repository — Online Retail II. Dataset page: https://archive.ics.uci.edu/dataset/502/online+retail+ii
Source note	The project used the local workbook. The public dataset URL is included as source attribution; verify the downloaded file/version matches the supplied workbook if exact provenance is required.

2. Dataset Purpose

The dataset records online retail transaction lines. It enables analysis of invoices, products, quantities, unit prices, transaction dates, customer identifiers, and countries. In this project it is used to build a repeatable ETL workflow and to support database-backed sales, customer, and product exploration.

3. Source Workbook Structure

The workbook contains two worksheets, which are combined by the ETL process:

Worksheet	Profiling information
Year 2009-2010	525,461 records reported during initial profiling.
Year 2010-2011	Combined with the first worksheet for the full extraction. The exact individual row count was not included in the retained project notes.
Combined extraction	1,067,371 records reported by the ETL run.

4. Raw and Processed Schema

Field	Meaning / use
Invoice / invoice_no	Invoice identifier; normalized to invoice_no in the processed schema.
StockCode / stock_code	Product or stock identifier.
Description / description	Product description text.
Quantity / quantity	Number of units on a transaction row; negative values may represent returns or adjustments.
InvoiceDate / invoice_date	Transaction date and time.
Price / unit_price	Unit price from the source workbook.
Customer ID / customer_id	Customer identifier; missing values occur in the source.
Country / country	Country associated with the transaction.
source_sheet	Added by ETL to preserve the originating worksheet.
transaction_value	Derived row-level transaction value, based on ETL transformation logic.
is_cancelled_invoice	Derived flag for cancelled invoices.

Field	Meaning / use
is_negative_quantity	Derived flag for negative quantity rows.

The processed schema has 12 columns. Exact transformation expressions and cancellation-identification rules are defined by the project source code and should be consulted when reproducing results.

5. Preprocessing Performed

The reported pipeline performs extraction, validation, transformation, cleaning, data-quality reporting, Parquet output, and MySQL loading. The combined dataset is normalized to the processed field names shown above. Derived transaction and status flags are added. Duplicate rows and anomalous values are measured and handled according to the implemented cleaning logic.

The project should not describe every negative quantity as invalid: retail datasets can use negative quantities to represent returns or corrections. Likewise, missing customer identifiers can still leave a transaction useful for sales totals while excluding it from customer-level analysis.

6. Data Quality Profile

Quality check	Reported count before cleaning
Total extracted records	1,067,371
Duplicate records	12,133
Negative quantity	22,950
Zero quantity	0
Negative price	5
Zero price	6,202
Invalid dates	0
Missing descriptions	4,382
Missing customer IDs	243,007

7. Processed Dataset and Database Results

Metric	Reported result
Cleaned dataset	1,055,233 rows × 12 columns
Processed file	data/processed/cleaned_retail.parquet
MySQL database	retail_analytics
Main table	transactions
Rows loaded to MySQL	1,055,233
Unique invoices	53,623
Unique products	5,131
Unique customers	5,942
Unique countries	43
Date range	2009-12-01 07:45:00 to 2011-12-09 12:50:00
Total quantity	10,575,682

Metric	Reported result
Total transaction value	19,390,476.658

The aggregate values are the results previously reported from the project's ETL and MySQL checks. Currency is not explicitly asserted here because the project notes do not establish a validated currency convention for all records.

8. Limitations and Interpretation Notes

• The record count falls from 1,067,371 to 1,055,233, a difference of 12,138 records. The data-quality report separately lists 12,133 duplicates, so the complete difference should be reconciled against the implemented cleaning code.

• Missing customer IDs affect customer-level metrics and can result in fewer identifiable customers than transaction rows.

• Negative quantities, cancellations, and zero-price rows may reflect legitimate business events; their interpretation depends on the transaction and cleaning rules.

• The exact row count for the second worksheet and the exact source-file download version were not present in the project notes used to prepare this document.

• The processed file and raw workbook are local artifacts. Keep large datasets out of GitHub unless the submission explicitly requires a permitted sample dataset. Include a small, non-sensitive sample if needed for evaluation.

9. Reproducibility

From the project root, ensure the workbook exists at data/raw/Online_Retail_II.xlsx, activate the Python environment, install requirements.txt, then run:

```
python run_etl.py
```

Expected successful extraction count reported during development: 1,067,371. Expected cleaned output shape: 1,055,233 rows and 12 columns. Actual counts may differ if the input workbook or code changes.

10. References

UCI Machine Learning Repository, Online Retail II: https://archive.ics.uci.edu/dataset/502/online+retail+ii

Project README: repository root README.md. ETL source: run_etl.py and modules under src/etl/. Database verification: project database test and SQL checks.